■ PLANRS ACADEMY — LESSON PLAN

Circuits: Buzzer Musical Keyboard!

Science — Diffusion, Colour & Pigments

■ **Course**	Circuits: Buzzer Musical Keyboard
■ **Grade**	Grade 3 – 6 (Ages 8–12)
■ **Duration**	30 Minutes
■ **Session**	Session 1 of 1
■ **Level**	Beginner
■ **Subject**	Science (+ Art & Music cross-links)
■■ **Context**	India-first examples throughout

What will students build today?

A working musical keyboard using a 9V battery, aluminium foil strips, and a piezoelectric buzzer — then play Sa Re Ga Ma Pa on it! Along the way, students discover the science of electric circuits, diffusion, and colour pigments — all through Indian everyday examples.

SECTION 1 — LEARNING OBJECTIVES

By the end of this 30-minute session, every student will be able to:

KNOW

Define a circuit, diffusion, and pigment in their own words. Name at least 3 circuit components: battery, wire, buzzer.

UNDERSTAND

Explain why a circuit must be closed to work. Describe diffusion using agarbatti or chai aroma. Explain how Holi gulal mixing creates new colours (subtractive).

DO

Assemble a working buzzer keyboard step-by-step. Produce sound by pressing foil keys to complete the circuit. Conduct a food-colour diffusion observation in water.

CONNECT

Link diffusion to how sound travels through air. Connect pigments to Indian fabric dyeing, rangoli, and mehndi.

Bloom's Taxonomy Coverage: *Remember → Understand → Apply → Analyse*

SECTION 2 — KEY CONCEPTS WITH INDIAN EXAMPLES

1 ELECTRIC CIRCUITS

A circuit is a closed path through which electricity flows to power a device. It needs three things: a power source (battery), a conductor (wire/foil), and an output device (buzzer or LED).

Indian Example: Doorbell (ghar ki ghanti) = battery + wire + buzzer!

2 OPEN vs. CLOSED CIRCUIT

Closed circuit = electricity flows and the device works. Open circuit = path is broken, electricity stops. Just like a road with a blockage — traffic (electricity) cannot move.

Indian Example: Diwali string lights: if one bulb breaks, the whole string goes dark (series circuit).

3 DIFFUSION

Diffusion is the movement of particles from a region of high concentration to low concentration until they are evenly spread. Particles move randomly — this spreading is the natural result.

Indian Example: Agarbatti lit in one corner — soon the whole room smells! Gulal thrown in air spreads everywhere.

4 COLOUR & PIGMENTS

A pigment is a substance that absorbs certain light colours and reflects others back to our eyes. In pigment mixing (subtractive): Red + Yellow = Orange. Blue + Yellow = Green.

Indian Example: Haldi (turmeric) is yellow because it reflects yellow light. Holi gulal = pigment mixing!

5 SOUND & THE BUZZER

A piezoelectric buzzer converts electrical energy into sound energy by vibrating rapidly. Sound travels through air by diffusion of energy — particles carry vibration from source to ear.

Indian Example: School ghanti (bell) = buzzer circuit. Sa Re Ga Ma = harmonium notes = different frequencies!

■ SECTION 3 — 30-MINUTE SESSION OUTLINE

Time	Segment	Activity
0–3 min	**■ HOOK**	Play a pre-built buzzer keyboard. Ask: 'Made from a battery, foil & buzzer — can you gu
3–6 min	**■ CONTEXT**	Circuits are everywhere: phone, TV remote, doorbell, auto horn. Introduce 3 science top
6–11 min	**■ CIRCUITS**	Open vs. closed circuit. Merry-go-round analogy. Diwali lights example. Quick Check ca
11–16 min	**■ DIFFUSION**	Agarbatti demo or food colour in water. Chai smell, petrol pump, Holi gulal examples. Po
16–20 min	**■ COLOUR & PIGMENTS**	What is a pigment? Haldi, mehndi, rangoli. Gulal mixing = subtractive colour. Colour wh
20–26 min	**■ BUILD ACTIVITY**	Step-by-step buzzer keyboard build: cut foil, connect battery, press keys, play Sa Re Ga
26–30 min	**■ WRAP UP**	3-2-1 Exit: 3 things learned, 2 surprises, 1 question. Students hold up their keyboards!

■ ***Instructor Tip:*** *Never lecture more than 3 minutes without an engagement trigger — a question, a poll, or a visual change. Every student in Grade 3–6 needs to be seen and heard.*

■ SECTION 4 — MATERIALS & PREPARATION

What You Need to Build the Buzzer Keyboard

Item	Quantity	Where to Find
9V Battery	1	Local electronics shop / kit
Piezoelectric Buzzer	1	Local electronics shop (■10–■30)
Aluminium Foil	5–7 strips	Kitchen (any home has this!)
Cardboard / Thick Paper	1 piece (A4+)	Old box or craft paper
Clear Tape / Cellotape	Sufficient	Any stationery shop
Scissors	1 pair	Have an adult help with cutting
Coloured Markers (Optional)	Set	Decorate your keyboard!

For Colour & Pigment Activity (Optional Add-On)

1 glass of water	2–3 drops of food colour (any colour)
Watercolour paints + paper	Haldi, neel (indigo), or mehndi powder

■ SAFETY FIRST!

- Use ONLY batteries under 9V. Never use wall sockets — they are dangerous!
- Always do this activity with a parent, guardian, or teacher nearby.
- Keep water away from the battery and all circuit components.
- Handle scissors carefully — ask an adult to help with cutting steps.
- If the buzzer gets very warm, disconnect the battery immediately.

SECTION 5 — BUILD IT! STEP-BY-STEP KEYBOARD GUIDE

Follow these steps carefully. After each step, check: 'Does it look right?' before moving on. Have an adult nearby at all times!

Step 1 PREPARE THE BASE

- Take a piece of cardboard (at least A4 size).
- This is the body of your keyboard — make it strong!
- Ask an adult to trim the edges neatly with scissors.
- Draw a rectangle outline where your 5 'keys' will go.

Step 2 CUT & PLACE THE FOIL KEYS

- Cut 5 strips of aluminium foil — each about 3 cm wide x 10 cm long.
- Tape them onto the cardboard, side by side, with a small gap between each.
- These foil strips are your piano keys — smooth them flat!
- Label them: Sa Re Ga Ma Pa (or 1, 2, 3, 4, 5).

Step 3 CONNECT THE RED WIRE (+)

- Take the red wire from your 9V battery connector.
- Connect (tape or clip) it to ONE terminal of the piezo buzzer.
- Red = Positive (+). Remember this always!
- Make sure the connection is firm — no loose ends.

Step 4 CONNECT THE BLACK WIRE (–)

- Take the black wire from the battery connector.
- Run this wire to the edge of your cardboard near the foil strips.
- Black = Negative (–). This wire will touch each key to complete the circuit.
- Leave enough wire length to comfortably reach all 5 keys.

Step 5 CONNECT THE BUZZER'S SECOND WIRE

- Take the other wire from the piezo buzzer.
- Connect it to the FIRST foil strip (Key 1 = 'Sa').
- When you press Key 1 AND touch the black wire to it — the circuit closes!
- You should hear a BUZZ! If not, check your connections.

Step 6 TEST ALL 5 KEYS

- Now connect the buzzer's output wire to each foil strip in turn.
- Each strip (different length of foil) may produce a slightly different tone.
- Press firmly — your finger helps complete the connection!
- Try playing: Sa Re Ga Ma Pa in sequence. Bahut badiya!

Step 7 DECORATE YOUR KEYBOARD (Bonus!)

- Use markers to colour-code your keys with Holi-inspired colours!
- Sa = Orange, Re = Yellow, Ga = Green, Ma = Blue, Pa = Purple.
- Add your name as the 'keyboard designer'.
- Show it to your family and play a tune!

Troubleshooting

No sound? Check: (1) Is the circuit closed — are all connections touching? (2) Is the battery new and charged? (3) Is the buzzer connected correctly — red to +? (4) Are the foil strips flat and making good contact? Try pressing harder on the key!

■ SECTION 6 — CONCEPT DEEP-DIVES WITH INDIAN EXAMPLES

■ Diffusion — The Agarbatti Science

Definition: Diffusion is the process by which particles move from a region of **high concentration** (many particles) to a region of **low concentration** (fewer particles), until they are evenly spread throughout the space.

■ Agarbatti smoke	Incense lit in one corner → smell fills entire room
■ Chai aroma	Brewing in kitchen → smells from the bedroom!
■ Holi gulal	Powder thrown in air → colour cloud spreads in all directions
■ Petrol pump smell	You smell petrol before you even reach the pump
■ Food colour in water	Drop of colour spreads through glass without stirring

■ **Analogy for Students**

Imagine 100 cricket balls packed tightly into one corner of a large box. Shake the box and the balls spread out to fill all the space equally. That is EXACTLY what particles do in diffusion —

■ Colour & Pigments — The Holi Colour Lab

Definition: A **pigment** is a substance that gives colour to objects by **absorbing** certain wavelengths of light and **reflecting** others back to our eyes.

Indian Pigment	Colour	Why?
Haldi (Turmeric)	Bright Yellow	Reflects yellow light, absorbs all others
Mehndi (Henna)	Green-Brown	Plant pigment from henna leaves
Gulal (Holi Red)	Red / Pink	Synthetic dye pigment
Neel (Indigo)	Deep Blue	Natural plant pigment, used in fabric dyeing
Kumkum	Deep Red	Pigment from turmeric + lime powder

Pigment Colour Mixing Chart (Subtractive): Pigment mixing is SUBTRACTIVE — more pigments = more light absorbed = darker colour. Opposite of RGB light mixing!

■ Red + ■ Yellow	=	**■ Orange**
■ Blue + ■ Yellow	=	**■ Green**
■ Red + ■ Blue	=	**■ Purple**
■ Red + ■ Blue + ■ Yellow	=	**■ Dark Brown / Black**

■ SECTION 7 — EXIT QUIZ (5 Questions)

Use these questions as a 5-minute exit assessment at the end of the session. Correct answers are highlighted in green with a brief explanation for the instructor.

Q1 **Which of the following BEST describes a closed circuit?**

(a) The battery is not connected to the buzzer

(b) Electricity flows through a complete, unbroken path ✓

(c) The wire is made of rubber

(d) The switch is turned off

Why: A closed circuit = complete path so electricity can flow continuously.

Q2 **Riya lights an agarbatti in her bedroom. Her grandmother in the living room smells it 5 minutes later. Which process explains this?**

(a) Evaporation

(b) Conduction

(c) Diffusion ✓

(d) Reflection

Why: Diffusion = particles moving from high to low concentration through air.

Q3 **When you mix RED gulal powder with YELLOW gulal powder during Holi, what colour do you get?**

(a) Purple

(b) Orange ✓

(c) Green

(d) Blue

Why: Red + Yellow pigments = Orange (subtractive colour mixing).

Q4 **What is the job of a piezoelectric buzzer in a circuit?**

(a) To store electricity

(b) To block electricity from flowing

(c) To convert electricity into sound ✓

(d) To measure voltage

Why: A buzzer is an output device — it converts electrical energy to sound energy.

Q5 **Which of these is the BEST conductor of electricity for your keyboard?**

(a) Rubber eraser

(b) Wooden ruler

(c) Aluminium foil strip ✓

(d) Plastic bottle cap

Why: Metals (like aluminium foil) are conductors — they let electricity flow freely.

■ SECTION 8 — SUCCESS CRITERIA

For Students
✔ 80%+ can define a circuit in their own words
✔ 75%+ cite agarbatti / gulal when asked for a diffusion example
✔ 70%+ have a working buzzer that makes sound
✔ Students use the word 'pigment' correctly in a sentence
✔ Students feel excited and confident about science (feedback poll)

For Content Quality
✔ All 4 Learning Objectives (LO1–LO4) addressed in the lesson
✔ Every concept has minimum 2 Indian context examples
✔ Worksheet follows easy → medium → challenge structure
✔ Assessment covers all 3 key concept areas
✔ Zero factual errors — reviewed by a subject matter expert

For Engagement
✔ Average session watch time ≥ 85% (recorded video format)
✔ 60%+ of students submit the exit quiz
✔ 40%+ students share buzzer keyboard photo or video
✔ Qualitative gold standard: 'My child talked about this at dinner!'

■■■ SECTION 9 — INSTRUCTOR QUICK REFERENCE CARD

Top 5 Indian Examples to Know Off by Heart

1	■ **Agarbatti spreading**	= Diffusion (smell particles in air)
2	■ **Holi gulal mixing**	= Pigment mixing (Red + Yellow = Orange)
3	■ **Diwali string lights**	= Series circuit (one breaks → all go off)
4	■ **Ghar ki ghanti**	= Buzzer circuit (button → circuit closes → DING!)
5	■ **Haldi / Turmeric**	= Natural yellow pigment (reflects yellow light)

Key Definitions — Say These Out Loud!

Circuit	A closed path through which electric current flows.
Diffusion	Movement of particles from high to low concentration.
Pigment	A substance that gives colour by absorbing and reflecting light.
Buzzer	A device that converts electrical energy into sound energy.
Conductor	A material that allows electricity to flow through it (e.g., metal).
Insulator	A material that does NOT allow electricity to flow (e.g., rubber, plastic).

Safety Reminder — Say This Before Build Activity

■■ **Read This Aloud Before Step 1**

"We are only using a 9-volt battery today — that is very safe. But please do NOT use any plug from the wall socket. Always have a parent or teacher with you. Let us build safely and smartly

Celebration Line — Use This at the End!

■ **Student Celebration Moment**

"Wah! You just built a musical keyboard using science! You understand circuits, diffusion, AND pigments now. You are officially a Circuit Scientist today. Bahut badiya! Ek baar aur bajao!"

■■ Things to Avoid

✗ Do NOT say 'electricity flows like water' without noting it is a simplification.

✗ Do NOT confuse pigment mixing (subtractive) with light mixing (RGB — additive).

✗ Do NOT use wall socket electricity for the demonstration — 9V battery only.

✗ Do NOT lecture more than 3 minutes without a question or engagement trigger.

✗ Do NOT use vocabulary above Grade 5 level without defining it immediately.

✗ Do NOT skip the safety disclaimer before the build activity.

✗ Do NOT use only North India examples — include South, East and West India references.

SECTION 10 — HOMEWORK & NEXT STEPS

Take-Home Experiment

Pigment Mixing at Home

Try mixing haldi (yellow turmeric) + neel (blue indigo) powder in a small bowl. What colour do you get? (Hint: it should be green!) Take a photo and share it in class next session. Can you predict what Red + Blue will make? Try it with watercolour paints!

Diffusion Observation

Light an agarbatti (with a parent's help) in one corner of your room. Stand at the opposite corner and note how many minutes it takes for the smell to reach you. Then try it on a hot day vs. a cool evening — is there a difference? Write down your observations!

Circuit Explorer Challenge

Walk around your home with a parent and find 5 devices that use circuits. Write them down: Name of device, what it outputs (light, sound, movement), and whether it has a switch to open/close the circuit. Share your list next session!

"Science is not a subject. It is the way the world actually works — and every child deserves to experience that magic."

— Planrs Academy Curriculum Philosophy

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